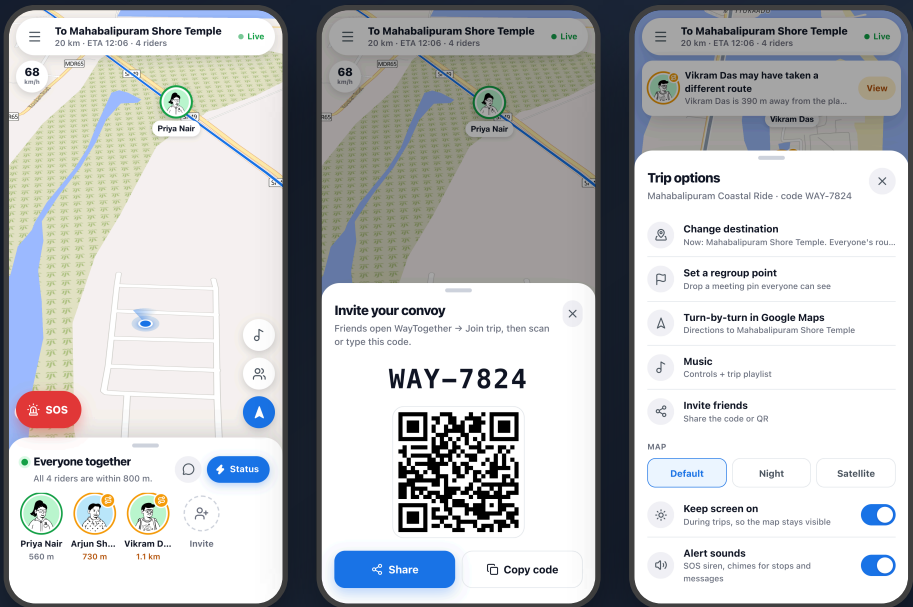




PRODUCTION KIT

Complete System Specification & Promotional Visual Kit

Real-time telemetry, 3D heading-up group navigation, instant QR convoy sharing, and automated road safety for connected travel groups.



Application Identifier
app.convoya.trip

Platforms
Android Native (Capacitor 8) · Modern Web PWA

Architecture
Zero-DB Ephemeral Supabase Realtime
Map Engine
MapLibre GL 3D (OpenFreeMap + Esri)

Version
v1.0.0 (Release-Ready)
Document Date
October 2026

01 Executive Summary & Product Vision

Document Directory

1. Executive Summary & Pillars	P. 02	2. Flagship: 3D Live Navigation	P. 03
3. Live Share & Convoy QR Join	P. 04	4. Convoy Health & Heuristics	P. 05
5. Automated Safety Matrix	P. 06	6. Emergency SOS & Siren System	P. 07
7. In-Ride Chat & Media Keys	P. 08	8. Onboarding & Route Setup	P. 09
9. Post-Trip Telemetry Logs	P. 10	10. Architecture & Tech Matrix	P. 11
11. Realtime Telemetry Schemas	P. 12	12. Native Android Subsystem	P. 13
13. Persistent Storage Contracts	P. 14	14. Build, APK & OTA Releases	P. 15
15. Promo Showcase (Part 1)	P. 16	16. Promo Showcase (Part 2)	P. 18
17. Promo Showcase (Part 3) & Specs	P. 20		

WayTogether (TRAVA) is a purpose-built real-time collaborative map and telemetry platform engineered specifically for groups traveling together — including motorcycle convoys, cycling clubs, car caravans, and overland exploration teams.

Conventional turn-by-turn navigation applications (such as Google Maps, Apple Maps, or Waze) are fundamentally architected around isolated individuals navigating from Origin A to Destination B. They fail to solve the critical challenges of collective movement on open highways:

3s

TELEMETRY INTERVAL

800m

PACK CLUSTER RADIUS

0 ms

SERVER DB LATENCY

55°

3D ROAD PITCH ANGLE

The Group Coordination Breakdown

When 4 to 20 riders embark on a highway trip, group cohesion naturally fragments. Riders get trapped behind heavy transport vehicles, take unexpected exits, or pull over due to mechanical failure or dehydration.

Today, convoys rely on messy WhatsApp location sharing (which throttles in the background, drains batteries, and lacks route context) or frantic roadside phone calls with riding gloves on.

The WayTogether Solution

WayTogether eliminates cognitive load on the rider. It continuously calculates the spatial spread of the entire group, projects riders along the shared road route in 3D perspective, detects stranded riders automatically, and provides glove-friendly quick communications.

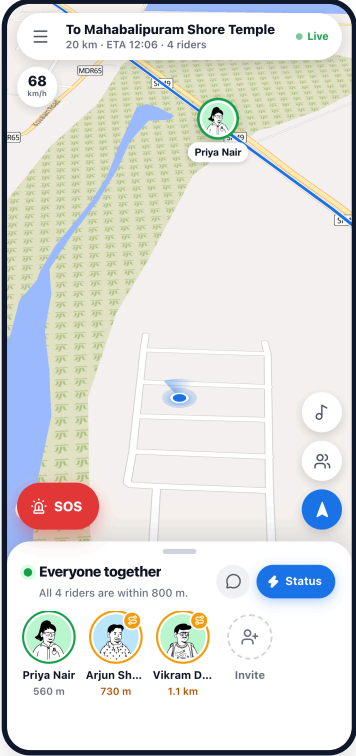
Best of all, it works ephemerally with zero database accounts required — just a 7-character trip code or QR scan.

Core Product Pillars

- 1. **Ephemeral Real-Time:** Zero SQL tables. Live synchronization operates over WebSocket channels (Broadcast + Presence).
- 2. **Background Endurance:** Continuous GPS broadcasts even when the phone is locked inside a jacket pocket or tank bag.
- 3. **Automated Convoy Health:** Heuristic engine actively watches for fallen-behind riders, stalled bikes, and route deviations.
- 4. **Distraction-Free Controls:** Hardware headset volume/media key mapping, one-tap status broadcasts, and synthesized sound chimes.

02 Flagship Feature: 3D Live Group Navigation

The core cockpit experience unites real-time 3D camera tracking with continuous multi-rider road positioning.



3D Live Group Navigation (Heading-Up Mode)

COCKPIT MODE

"True 3D perspective tracking your entire convoy on the road ahead."

When riding at highway speeds, a 2D top-down map forces the rider to mentally translate rotations. WayTogether's cockpit mode tilts the MapLibre GL camera to a **55° pitch** and automatically rotates bearing to match device heading, positioning the user low on the display:

- ✓ **Real-Time Road Perspective:** Convoy riders ahead appear physically further up the tilted highway route.
- ✓ **Floating Telemetry Badge:** High-contrast HUD speedometer (e.g. **68 km/h**) calibrated from high-precision GPS fixes.
- ✓ **Dynamic Trip Header:** Displays live remaining distance (**20 km**), ETA (**12:06**), and active rider count.
- ✓ **Convoy Proximity Pill:** Live green status indicator (*"Everyone together · All 4 riders are within 800 m"*).
- ✓ **Instant Rider Distance Tags:** Live distance badges directly below each avatar (*"Priya Nair 560 m"*, *"Vikram Das 1.1 km"*).
- ✓ **One-Tap SOS Access:** High-visibility red emergency button positioned for immediate thumb activation.

Camera Mode Matrix

- 'nav': 3D perspective, 55° pitch, heading-up rotation, offset [0, 110] for forward road preview.
- 'group': Dynamic 2D bounding box auto-fitting all convoy members and the destination route.
- 'me': Top-down locked tracking on user's current GPS position.
- 'free': Unlocked pan and zoom gesture exploration.

The image is a screenshot of a mobile application interface. The top portion shows a map with a blue route line. A location pin is placed on the map, labeled 'Priya Nair' with a profile picture icon. Above the map, text reads 'To Mahabalipuram Shore Temple', '20 km · ETA 12:06 · 4 riders', and a 'Live' status indicator. A speedometer graphic shows '68 km/h'. The bottom portion of the screen is a white overlay with the heading 'Invite your convoy'. Below this, it says 'Friends open WayTogether → Join trip, then scan or type this code.' The code 'WAY-7824' is displayed in large, bold, black letters. Below the code is a large QR code. At the bottom of the overlay are two buttons: a blue button with a share icon and the text 'Share', and a white button with a copy icon and the text 'Copy code'.

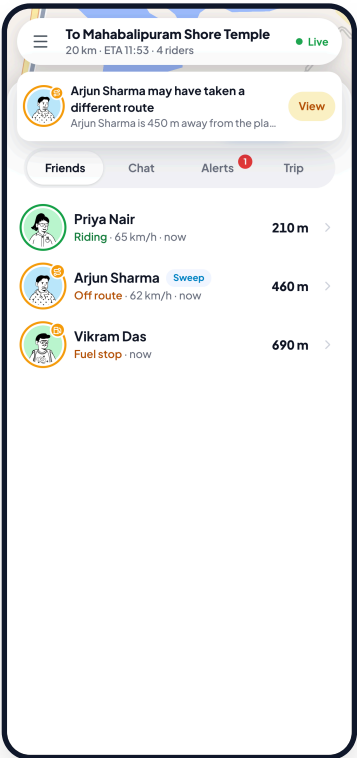
FRICTIONLESS JOIN

- ✓ **Prominent High-Contrast Code:** Giant **WAY-7824** badge clearly readable from 3 meters away across parked motorcycles.
- ✓ **High-Resolution Vector QR Code:** Friends open their app camera and join the room instantly without typing a single character.
- ✓ **Native App Share Sheet:** One-tap Share button triggers Android's system share intent to send direct join links via WhatsApp, Telegram, or SMS.
- ✓ **Clipboard Copy Action:** Quick copy button for posting to travel forums or chat channels.
- ✓ **Pre-Join Inspection (`LookupTrip`):** Peeks into the channel before full subscription, displaying trip name, destination, and current rider count so joining users confirm legitimacy.

Trip codes exist exclusively for the duration of the journey. Once the trip leader taps "End Trip", the WebSocket channel disconnects, all location broadcasts cease immediately, and no residual location trails remain on external servers.

04 Convoy Health Engine & Group Heuristics

WayTogether replaces manual group check-ins with an intelligent mathematical spatial evaluation engine running client-side (`TripContext.jsx`).

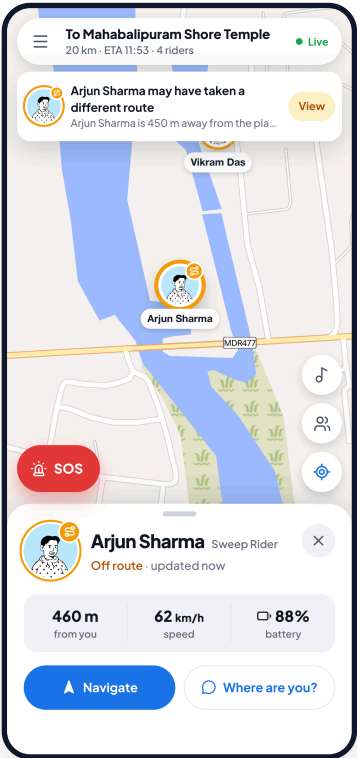


Algorithmic Convoy Clustering

Every incoming GPS coordinate fix is evaluated against the convoy's spatial cluster using the Haversine spherical formula and orthogonal route snapping:

Status	Condition	Visual & Audio Feedback
OK Together	All riders separated by $\leq 800\text{m}$ (or in groups ≥ 4 , max 1 within 1.5km)	Green pill: "Everyone together"
WARN Spreading	Any rider drifts between 1.5 km and 3.0 km from nearest member	Amber banner + gentle chime alert
DANGER Split	Any rider falls $> 3.0\text{ km}$ behind the convoy	High-priority red alert toast

Linear Convoy Strip: The bottom sheet projects all members onto a 1-dimensional route line, showing front-runners, the designated Sweep, and lagging riders with meter-precise gap indicators.



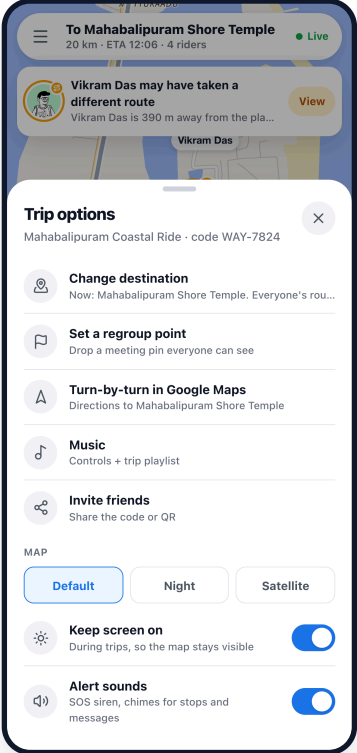
Individual Rider Telemetry & Direct Intercept

Tapping any rider on the map or roster pulls up deep real-time diagnostic telemetry:

- ✓ **Tri-Metric Real-Time Readout:** Distance from user (e.g. 460 m), current speed (62 km/h), and phone battery level (88%).
- ✓ **Activity Status:** Shows whether the friend is actively riding, stopped for fuel, having breakfast, or resting.
- ✓ **Direct Intercept Navigation:** One-tap "Navigate in Google Maps" button opens driving directions straight to their current GPS coordinate.
- ✓ **Quick Ping:** Sends an instant gentle vibration ping to their handlebars.

05 Automated Safety & Anomaly Detection

The safety engine actively monitors individual telemetry streams to detect mechanical breakdowns, navigational mistakes, and dropped riders.



Dynamic Routing & Off-Route Detection

LEADER & SAFETY

"Proactive notifications the moment a rider takes a wrong highway fork."

Highway journeys are fluid. Leaders need to adjust plans, and the system must alert when a convoy member takes an accidental bypass:

- ✓ **Off-Route Anomaly Alert:** Proactive notification banner (*"Vikram Das may have taken a different route — 390 m away"*) with a direct "View" camera snap button.
- ✓ **Mid-Trip Destination Change:** Leader updates destination on the fly; all member phones recalculate routes instantly.
- ✓ **Regroup Meeting Pins:** Drop a shared pin at a gas station or viewpoint visible to the entire group.
- ✓ **External Turn-by-Turn Handoff:** One-tap deep link launches Google Maps turn-by-turn navigation directly to destination.

Comprehensive Anomaly Detection Matrix

Trigger	Threshold / Condition	System Action	Severity
Not Moving (Still)	Rider moves < 40m for > 5 min without declaring stop	Alerts group: <i>"Arjun hasn't moved for 5 min"</i> . Prompts check-in.	WARNING
"Are You OK?" Check-In	Convoy member taps "Ask if OK"	Dispatches priority dialog to stopped rider with 2-minute countdown.	URGENT
Check-In Timeout	Rider fails to answer within 120 seconds	Auto-triggers SOS on their behalf; alerts all riders to navigate immediately.	EMERGENCY
Off-Route Deviation	Rider drifts > 300m perpendicular to OSRM route	Toasts group: <i>"Rider took a different route (390 m away)"</i> .	WARNING
Low Battery Alert	Device battery level drops < 15%	Broadcasts low battery warning so convoy knows location may freeze.	INFO
Auto-Resumption	Rider moves > 500m from a declared stop	Automatically switches user status from Fuel/Food back to riding.	SUCCESS

06 Safety & Emergency SOS System

Safety is non-negotiable on long highway rides. WayTogether integrates an industrial-grade emergency response mechanism engineered to operate under high-stress conditions.

Emergency Assistance

Press and hold for 3 seconds to alert your convoy.

SOS

Hold 3 sec

Shares your location, name, time and trip with 3 friends.

Call Emergency Services · 112

Deliberate 3–Second Hold Activation

LIFE SAFETY

"Prevents false triggers while enabling immediate emergency broadcasting."

Bumps and accidental screen touches in pockets must never fire emergency alarms. The SOS trigger requires a sustained **3–second continuous hold**:

✓ Continuous Haptic Feedback:

Phone vibrates with increasing intensity during the 3–second hold.

✓ Animated SVG Progress Ring:

Circular timer gives tactile and visual confirmation of arming.

✓ Full–Screen High–Contrast Red Interface:

Screen switches to high-visibility crimson (#B3121F) with elapsed emergency duration timer.

✓ Synthesized Dual–Tone Siren:

Browser Web AudioContext generates continuous 980 Hz / 660 Hz square-wave siren audio without relying on external MP3 downloads.

✓ Immediate Convoy Handoff:

Every member receives an urgent notification. Tapping it opens direct turn-by-turn routing to the distressed rider's exact latitude/longitude coordinates.

✓ One–Tap Emergency Services Dial:

Quick button to instantly dial national emergency dispatch (112).

Automated Check-In Escalation Chain

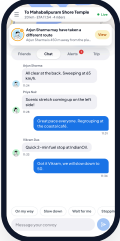
If a motorcycle rider slides off the road into a ditch and is incapacitated, they cannot press the SOS button. WayTogether's automated check-in engine detects the absence of movement (> 5 min), sends an "Are You OK?" push prompt with a 120–second audible countdown, and if unacknowledged, **automatically fires the SOS broadcast on the rider's behalf**, locking their coordinates and routing the group back to help.

Confidential · All Rights Reserved

Page 7 of 20

07 In-Ride Communication, Status & Hardware Controls

Operating a touchscreen while wearing riding gloves or driving is dangerous. WayTogether provides tactile, distraction-free coordination channels.



In-Convoy Chat

GLOVE-FRIENDLY

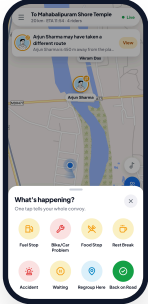
Ephemeral chat stored locally (up to 150 messages) without permanent servers.

8 Quick Chips:

ON MY WAY SLOW DOWN WAIT FOR ME

STOPPING SOON ALL GOOD WHERE ARE YOU?

TAKE NEXT EXIT LET'S REGROUP



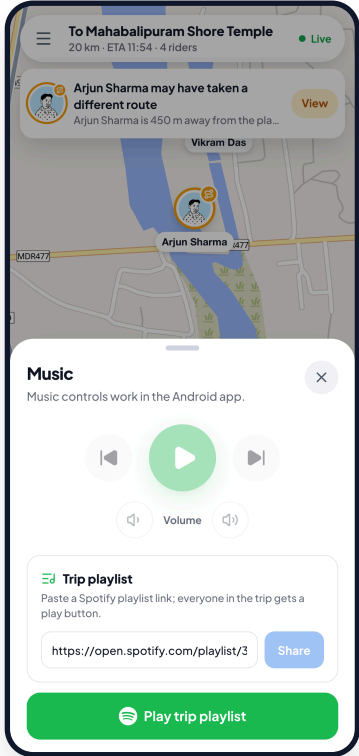
One-Tap Status Picker

BROADCAST

Instant broadcast of rider condition with visual map markers.

8 Distinct High-Contrast States:

- Fuel Stop (Orange badge & gas icon)
- Problem (Amber warning with motorcycle icon)
- Food Stop (Yellow fork/knife)
- Rest Break (Teal coffee mug icon)
- Accident (Red flashing alert icon)
- Regroup (Indigo flag icon)



Hardware Media Keys & Shared Spotify Playlist

NATIVE ANDROID

"Control music via handlebar remotes and sync group road playlists."

Long distance touring requires great soundtracks. WayTogether connects directly to the phone's native media framework:

- ✓ **Native Android Media Bridge:** Custom `MediaControlPlugin.java` dispatches hardware `KeyEvent` actions directly into Android's `AudioManager`.
- ✓ **Background Player Independence:** Works seamlessly whether the rider is playing Spotify, YouTube Music, Apple Music, or local MP3 files.
- ✓ **Handlebar Bluetooth Remotes:** Riders with Bluetooth handlebar buttons can trigger Play/Pause, Next Track, and Volume Adjust without leaving the navigation screen.
- ✓ **Synchronized Road Playlist:** Leaders can paste any Spotify collaborative playlist link; all members get a synchronized one-tap launch shortcut.

The image displays three sequential screens of the 'WayTogether' mobile application, illustrating its user flow from onboarding to trip planning.

Screen 1: Welcome & Onboarding

- Header:** 'WayTogether' logo and name.
- Main Text:** 'Keep the Journey Together.' followed by a description: 'Stay connected with your group. Track everyone, coordinate stops, and stay safer on every journey.'
- Features List:**
 - See every friend live on one map
 - Know who stopped, who's behind, who took a wrong turn
 - Hold SOS to alert your whole convoy
- Buttons:** 'Continue with Google' (with Google logo) and 'Continue without an account'.
- Footer:** A small disclaimer: 'Next, allow location. It's shared only with trips you join, until you leave.'

Screen 2: Home Dashboard

- Search Bar:** 'Where to?' with a location pin icon.
- Map:** A map showing a current location (blue dot) on 'South Mada Street'.
- Card: Ride together, Alex**
 - Text: 'See your friends live on one map, all trip long.'
 - Buttons: 'Start a trip' and 'Join trip'.
 - Last trip: Ooty Viewpoint**
 - Details: '84 km · 4h 0m · 2 trips total'.
 - Icon: A car icon.
 - Footer: 'Location is shared only while you're in a trip'.

Screen 3: Route & Convoy Setup

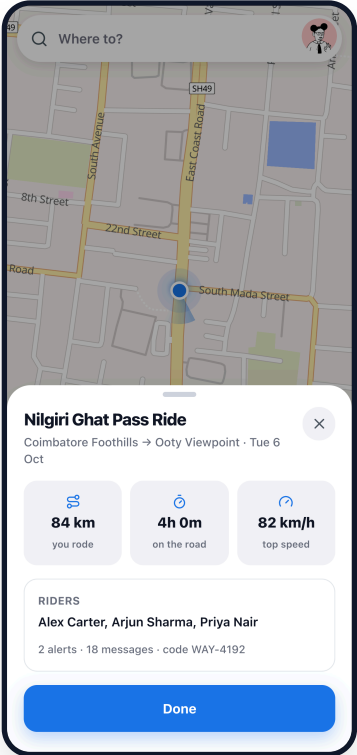
- Header:** 'Mahabalipuram' with a back arrow.
- Map:** A map showing a route from a starting point to 'Mahabalipuram'.
- Card: Mahabalipuram**
 - Location: 'Tamil Nadu, India'.
 - ETA/Distance: '39 min · 42 km'.
 - Text: 'Trip name'.
 - Input field: 'Mahabalipuram trip'.
 - Mode Selection: 'Bike', 'Car', and 'Mixed' (selected).
 - Button: 'Create group trip'.

Users can authenticate in one tap with Google (via `@capgo/capacitor-social-login`) or simply enter a nickname. DiceBear Notionists generates an instant, deterministic SVG avatar that syncs across all convoy phones.

Powered by Komoot Photon (OSM) search and Project OSRM driving calculations. Leaders select convoy type (**Bike**, **Car**, or **Mixed**) to calibrate speed limits and proximity calculations accordingly.

09 Post-Trip Summary & Telemetry Analytics

When the leader terminates a ride, all participants receive an interactive post-trip analytics dashboard celebrating their achievements.



Trip Milestone Recap & Logs

MILESTONE RECAP

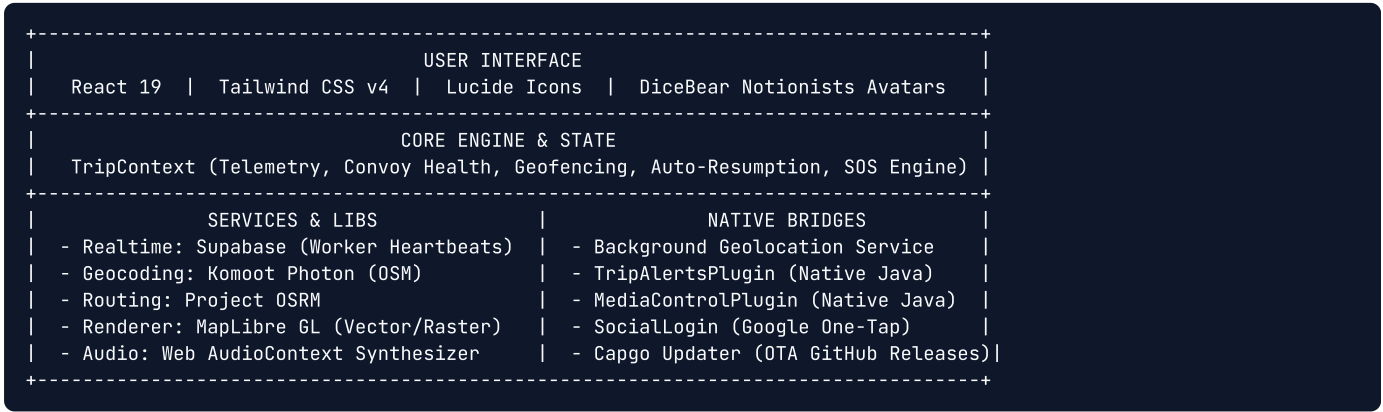
"Celebrate the ride with comprehensive telemetry stats and rider logs."

Every ride logs key telemetry metrics locally in device storage:

- ✓ **Distance Ridden:** Precise odometer tally (e.g. 84 km you rode).
- ✓ **Saddle Time:** Total active highway duration (e.g. 4h 0m on the road).
- ✓ **Speed Telemetry:** Safe top-speed record (e.g. 82 km/h top speed).
- ✓ **Convoy Roster Log:** Complete list of participants who completed the ride.
- ✓ **Event Counters:** Total safety alerts, regroupings, and chat messages exchanged during the journey.
- ✓ **Local Persistence:** Automatically saved in device history (up to last 30 trips) for offline bragging rights.

10 High-Level Architecture & Tech Matrix

WayTogether is built on a modern, decoupled client-first stack combining Web Standards with Native Android integrations.



Complete Technology Matrix

Layer	Technology	Version	Purpose & Technical Rationale
Frontend Framework	React	^19.3.0	Concurrent rendering, modern hooks, zero-latency reactive state.
Build Tooling	Vite	^8.3.3	Lightning-fast ES-module bundling, instant HMR, minimal bundle footprint.
CSS Framework	Tailwind CSS	^4.3.3	Utility-first styles with full safe-area insets for notched mobile displays.
Map Rendering	MapLibre GL	^6.13.0	GPU-accelerated vector rendering with 3D camera pitch and custom markers.
Map Tiles	OpenFreeMap & Esri	Free / Open	Vector tiles (Liberty/Dark) & Satellite raster imagery with no API keys.
Routing Engine	Project OSRM	Public API	Turn-by-turn routing with speed, geometry, distance, and duration.
Geocoding & Search	Komoot Photon	Public API	OpenStreetMap-based forward search and reverse geocoding.
Realtime Messaging	Supabase JS	^2.117.3	Web Worker-backed WebSocket connection for Broadcast & Presence.
Native Runtime	Capacitor	^8.5.3	Modern, lightweight web-to-native Android runtime bridge.
Background GPS	Capacitor Geolocation	^1.2.26	Persistent Android foreground service tracking locked phones.
Keep Awake	Capacitor KeepAwake	^8.0.1	Prevents screen sleep while phone is mounted on motorcycle handlebars.
Social Auth	Capgo Social Login	^8.5.12	Native Google Identity sign-in with Web Client ID.
OTA Updates	Capgo Updater	^8.52.1	Instant GitHub Releases Over-the-Air bundle updater.

11 Real-Time Protocol & Telemetry Schemas

WayTogether operates with a **Zero-Database Ephemeral Design**. Trip state lives exclusively in memory across participating devices via Supabase Realtime Channels:

Channel Name: waytogether-trip-{TRIP_CODE} (e.g. waytogether-trip-WAY-7824)

1. Broadcast Channel (High Frequency)

Transmits GPS fixes every ~3 seconds, instant status broadcasts, chat chips, and emergency SOS packets. No database read/write bottlenecks.
2. Presence Channel (State Sync)

Maintains active participant rosters and stores latest trip metadata (revision counter, destination, leader identity) for newly joining riders.

Location Packet Schema (event: 'loc')

```
{
  "id": "usr_9b1deb4d3b7d4",
  "name": "Priya Nair",
  "photo": "https://api.dicebear.com/9.x/notionists/svg?seed=Priya",
  "role": "leader",
  "lat": 12.825120,
  "lng": 80.241510,
  "acc": 6,
  "speed": 68,
  "heading": 175,
  "battery": 92,
  "status": "riding",
  "updatedAt": 1775626800000,
  "trip": {
    "code": "WAY-7824",
    "name": "Mahabalipuram Shore Temple Ride",
    "type": "Bike",
    "rev": 1,
    "dest": {
      "name": "Mahabalipuram Shore Temple",
      "latlng": [12.6166, 80.1994]
    }
  }
}
```

Event Packet Schema (event: 'evt')

```
// 1. Group Chat Message
{ "type": "msg", "msg": { "id": "m1", "from": "usr_1", "name": "Priya", "text": "Fuel stop in 2 km", "at": 1775626815000 } }

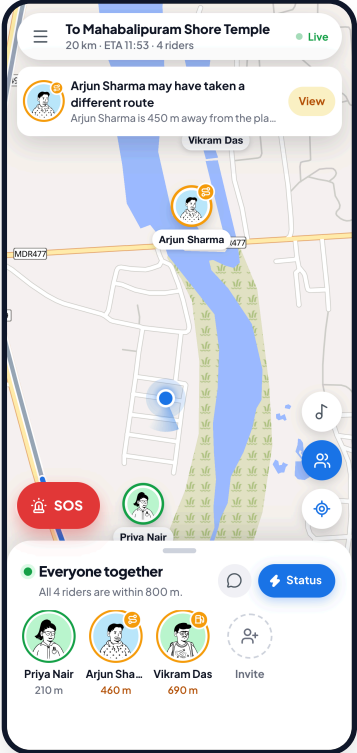
// 2. Check-In Prompt & Reply
{ "type": "checkin", "id": "chk_01", "to": "usr_2", "from": "usr_1", "fromName": "Priya" }
{ "type": "checkin-reply", "id": "chk_01", "from": "usr_2", "fromName": "Arjun", "answer": "stop", "label": "Fuel stop" }

// 3. Regroup Point Broadcast
{ "type": "regroup", "latlng": [12.7541, 80.2215], "label": "Coffee Day Highway Regroup" }

// 4. Trip Termination (Leader Only)
{ "type": "end", "by": "usr_1", "byName": "Priya Nair" }
```

12 Native Android Subsystem & Background Location

To guarantee continuous tracking when phones are locked inside motorcycle jackets, WayTogether integrates deep Android OS services via Capacitor.



Foreground Location Service

Android terminates normal background background apps after minutes to save battery. WayTogether launches an official **Android Foreground Service** ([@capacitor-community/background-geolocation](#)):

- ✓ **Persistent Status Bar Notification:** Displays "Sharing your live location · Your convoy can see where you are" with zero dismissal risk.
- ✓ **5-Meter Distance Filter:** Batches coordinate fixes dynamically to balance meter-accuracy with multi-hour battery longevity.
- ✓ **Web Worker WebSocket Heartbeats:** Supabase client is initialized with `{ realtime: { worker: true } }` so WebSocket pings execute uninterrupted in a dedicated thread.

Custom Native Android Plugins

Native Plugin	Source Location	Responsibilities & Implementation Details
TripAlertsPlugin.java	android/app/.../TripAlertsPlugin.java	Direct <code>NotificationManagerCompat</code> wrapper bypassing Android 14+ exact alarm blocks. • urgent channel: High importance, alarm category, custom vibration pattern <code>[0, 400, 200, 400, 200, 800]</code> . • trip channel: High importance, message category for off-route alerts. • chat channel: Default importance for group chat notifications.
MediaControlPlugin.java	android/app/.../MediaControlPlugin.java	Accesses Android <code>AudioManager</code> to dispatch native <code>KeyEvent</code> actions (<code>KEYCODE_MEDIA_PLAY_PAUSE</code> , <code>KEYCODE_MEDIA_NEXT</code> , <code>KEYCODE_MEDIA_PREVIOUS</code>) into active background media players.

13 Persistent Storage Contracts & State Lifecycle

All client-side state is strictly namespaced under `convoya.*` to ensure modular cleanup and conflict avoidance:

Storage Key	Format	Retention	Purpose & State Schema
<code>convoya.profile</code>	JSON Object	Permanent	User identity: <code>{ id, name, photo, email, google: true }</code> .
<code>convoya.trip</code>	JSON Object	Active Trip	Active trip state: <code>{ code, role, meta, status, stats }</code> . Allows resumption if app restarts.
<code>convoya.chat.<CODE></code>	JSON Array	Active Trip	Local message history up to 150 items: <code>[{ id, from, name, text, at }]</code> .
<code>convoya.history</code>	JSON Array	Permanent	Last 30 completed trip summaries with distance, duration, top speed, and rider lists.
<code>convoya.settings</code>	JSON Object	Permanent	Preferences: <code>{ mapStyle: 'default' 'night' 'satellite', sounds: true, keepAwake: true }</code> .
<code>convoya.update</code>	JSON Object	Ephemeral	Metadata for pending Over-The-Air bundle: <code>{ build, bundleId }</code> .

Crash Resumption Protocol

If a rider's phone restarts, experiences an Android OS kill, or the battery dies and recharges, WayTogether performs an automatic zero-click session restore:

- Inspects `convoya.trip` on boot. If an active trip exists, immediately bypasses Home and reconnects to `waytogether-trip-{CODE}`.
- Restarts the Background Geolocation foreground service without prompting the user.
- Broadcasts a re-announcement `Loc` packet so other riders see the user reappear online without missing a beat.

14 Build, Deployment & OTA Update Architecture

WayTogether features a dual-tier release pipeline separating agile frontend updates from native APK releases.

1. Web Bundle OTA Updates

UI features, map styles, and logic changes deploy Over-the-Air without requiring users to reinstall the APK:

```
# 1. Push code to main branch
git push origin main

# 2. GitHub Actions runs publish-update.yml:
# - Builds Vite production bundle
# - Computes SHA-256 hash
# - Generates update.json manifest
# - Creates GitHub Release with bundle.zip
```

App polls manifest every 5 minutes in background and prompts restart on Home screen.

2. Native APK Build (Android Studio)

Required only when adding new native Capacitor plugins, permissions, or Java code:

```
# 1. Build Vite web bundle
npm run build

# 2. Sync to Android project
npx cap sync android

# 3. Compile APK with JDK 21
cd android
JAVA_HOME=/path/to/jdk-21 ./gradlew assembleDebug

# Output: android/app/build/outputs/apk/debug/app-debug.apk
```

Local Convoy Multi-Rider Testing

Developers can simulate an entire 4-rider convoy on a single development machine without real GPS movement:

```
# Leader:
http://localhost:5173/?mock=12.9830,80.2594&heading=180&speed=45&local=1

# Member 1 (Trailing by 500m):
http://127.0.0.1:5173/?mock=12.9875,80.2590&heading=180&speed=42&local=1

# Member 2 (Off-Route Simulation):
http://localhost:5173/?mock=12.9840,80.2650&heading=140&speed=35&local=1

# Rapid Anomaly Testing Overrides:
?stillMin=0.5      # Accelerates "Not Moving" alert from 5 min to 30 sec
?checkinSec=20    # Accelerates "Are You OK?" timeout from 120s to 20s
```

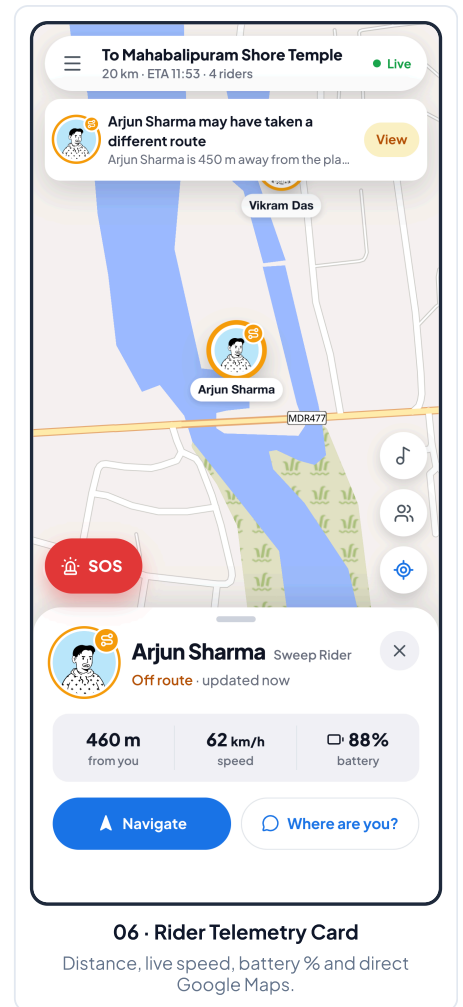
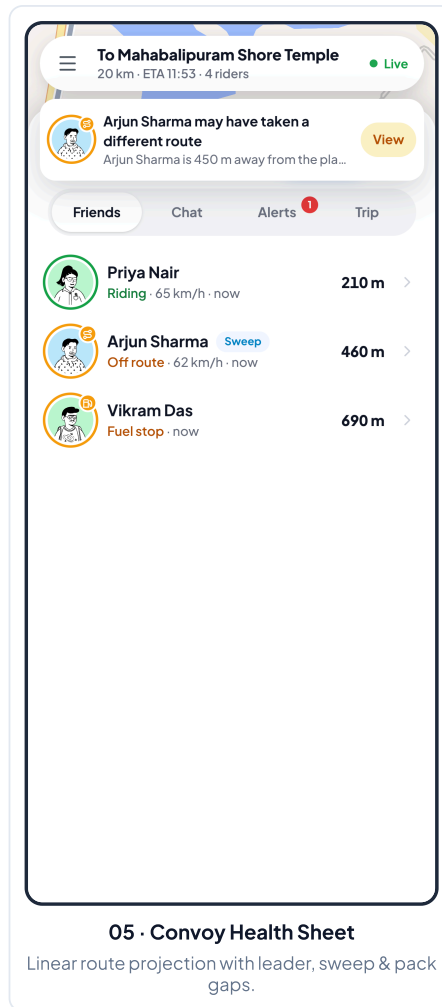
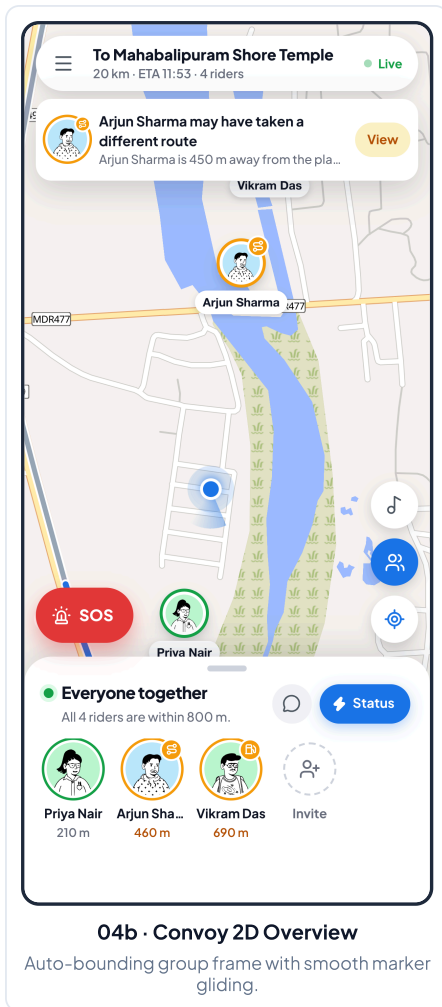
15 Complete 14–Screen Promotional Showcase (1/3)

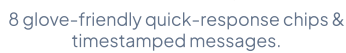
All screenshots are rendered at 1236 × 2676 px (3× Retina mobile density) and preserved in lossless 24–bit PNG format at TRAVA/promo-screenshots/ :

04 · 3D Cockpit Navigation
55° heading-up view with live speed badge & distance tags.

12 · Live Share & QR Code
Prominent 7–char code with sharp scannable QR sheet.

13 · Route Controls & Anomaly
Off-route toast notification & regroup meeting pins.





Emergency Assistance

Press and hold for 3 seconds to alert your convoy.

SOS

Hold 3 sec

Shares your location, name, time and trip with 3 friends.

Call Emergency Services · 112

10 · Emergency SOS System

3s hold activation, Web Audio siren & coordinated broadcast.

Where to?

Nilgiri Ghat Pass Ride

Coimbatore Foothills → Ooty Viewpoint · Tue 6 Oct

84 km

you rode

4h 0m

on the road

82 km/h

top speed

RIDERS

Alex Carter, Arjun Sharma, Priya Nair

2 alerts · 18 messages · code WAY-4192

Done

11 · Post-Trip Analytics

Distance ridden, saddle time, top speed & member log.

WayTogether

Keep the Journey Together.

Stay connected with your group. Track everyone, coordinate stops, and stay safer on every journey.

See every friend live on one map

Know who stopped, who's behind, who took a wrong turn

Hold SOS to alert your whole convoy

Continue with Google

Continue without an account

Next, allow location. It's shared only with trips you join, until you leave.

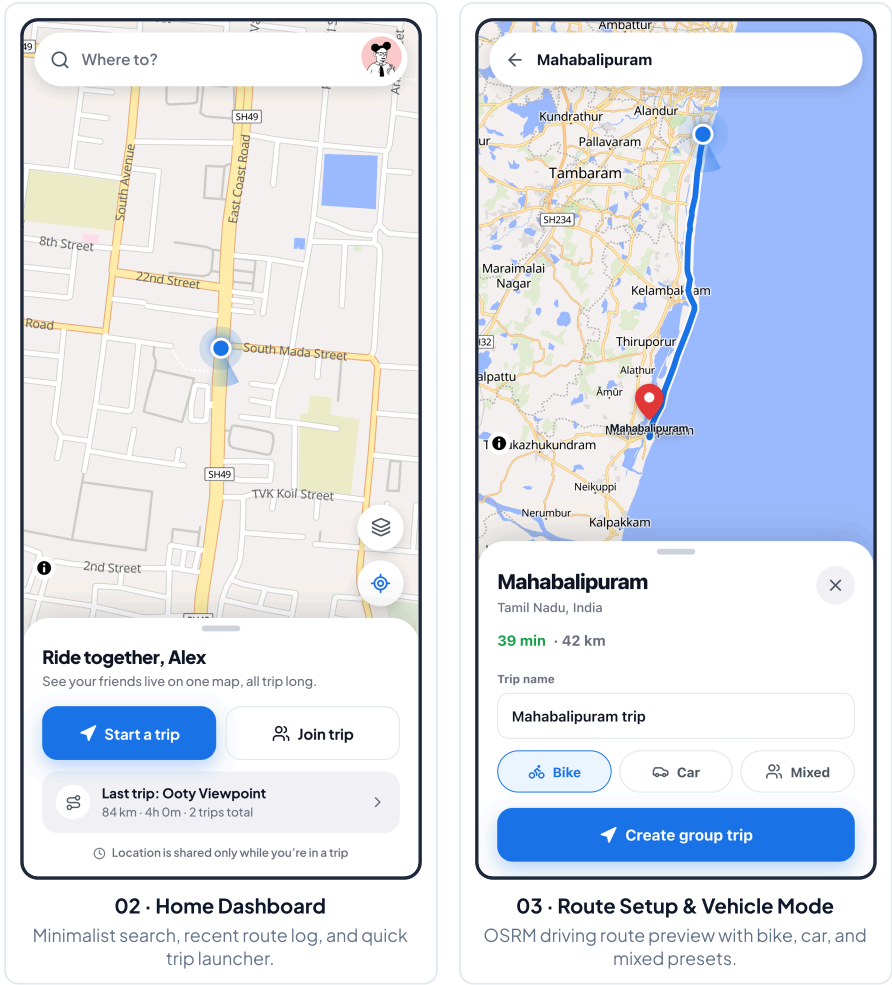
01 · Welcome & Onboarding

Clean brand identity with Google One-Tap & guest sign-in.

Confidential · All Rights Reserved

Page 19 of 20

17 Complete Promotional Showcase (3/3) & Specs



Asset Specifications & Campaign Applications

Specification	Metric / Value	Recommended Marketing Channel
Pixel Dimensions	1236 × 2676 px	Google Play Store Feature Carousel & Apple App Store 6.7" Mockups
Pixel Density	3× Retina Scale	High-DPI Investor Pitch Decks & Printed Brochures
Aspect Ratio	~9:19.5 (Flagship Mobile)	Instagram / TikTok Video Reels & Product One-Pagers
Color Profile	sRGB (Lossless 24-bit PNG)	Web Landing Page Hero Sections & Interactive Product Tours